Surya Deip Reddy Kesaram

United States of America • (513) 616-3315 • suryadeipreddy@gmail.com • linkedin.com/in/suryadeipreddy/ • https://github.com/suryadeipreddyk

Summary

• Software Engineer with 2.5+ years of experience designing and building scalable backend services and cloud-native applications on AWS, Azure, and GCP.
• Proficient in Python, Java, and SQL, with expertise in RESTful APIs, data integration, and automating infrastructure using modern DevOps practices.
• Demonstrated ability to reduce infrastructure costs by 20%, automate manual processes by 40% through efficient cloud architecture and CI/CD workflows.
• Skilled in full-lifecycle software development, using Agile principles to build robust applications at the intersection of Data, Cloud, and Backend systems.

Work Experience

Software Engineer Intern, *University of Dayton Information Technologies, USA* — **August 2024 – August 2025**

• Designed and developed backend services to integrate student, faculty, and staff data from departments, streamlining reliability across university records.
• Built Python and SQL based backend modules for transforming and securely handling sensitive data, ensuring data quality, privacy with FERPA regulations.
• Developed backend services on AWS to handle REST API requests for course enrollment, processing and updating student records on the university portal.
• Orchestrated workflows using Jenkins to automate the new data into the institutional data platform, reducing manual intervention by 40%.
• Integrated service center data from TeamDynamix APIs into internal applications, enabling real-time analytics into IT support and ticketing metrics.
• Participated in Agile sprints with developers, gathering requirements and delivering features for the university's GPT to integrate academic needs.

Graduate Research Assistant, *University of Dayton, USA* — **August 2023 - August 2024**

• Conducted extensive research on diverse datasets (structured, unstructured, and semi-structured) for AI/ML, Image Processing, and Data Engineering.
• Built ETL prototypes to extract, transform, and load research datasets into PostgreSQL and cloud storage environments for downstream analysis.
• Wrote Python scripts and SQL queries to transform and standardize raw data, improving quality and consistency across academic and research projects.
• Coordinated on data modelling design and schema optimization to support efficient data access and retrieval in OLTP/OLAP warehousing environments.
• Collaborated with a team of five researchers to develop a web-based JSON API application that enhanced data extraction efficiency by 30%.
• Developed and implemented ML models for predicting house prices and assessing weather-related risks impacting property values for users.

Institutional Metadata Engineer, *University of Dayton Libraries, USA* — **April 2024 – May 2024**

• Migrated 1,000+ scholarly blog articles to eCommons, by transforming content into structured metadata for seamless ingestion into Bepress.
• Enhanced metadata quality by conducting thorough quality assurance checks, aligning records with archival and metadata standards
• Synchronized with digital and archival teams to implement strategies for migration, metadata preservation.

AWS Cloud Architect Intern, *Amazon, India*. — **October 2021 – December 2021**

• Designed and deployed a scalable, cloud architecture for hosting a coffee house web app using AWS EC2, Auto Scaling, S3, SNS, API Gateway, and Lambda
• Optimized AWS EC2 instances for cost efficiency by utilizing reserved instances and auto-scaling, reducing infrastructure costs by 20%.
• Devised a web interface for Auto Scaling Groups—creation, policy definition, resource monitoring, SNS alerts—90%+ user satisfaction, 95% app uptime.
• Collaborated with DevOps teams to integrate AWS services with CI/CD pipelines, improving deployment speed and reliability.
• Strengthened AWS proficiency by gaining end-to-end experience in infrastructure setup, monitoring, automation, and cost-effective scaling.

Cyber Security Architect Intern, *Cisco Networking Academy, India* — **March 2021 – June 2021**

• Envisioned and simulated university-wide network infrastructure using Cisco Packet Tracer, apply advanced topologies and layered security protocols.
• Established secure network architectures and implemented cybersecurity protocols across complex topologies.
• Conducted network security audits and risk assessments to identify and mitigate vulnerabilities in network configurations, applying resolution strategies.
• Enhanced hands-on cybersecurity skills by addressing real-world scenarios and ensuring high standards of data protection and network reliability.
• Collaborated in Agile teams and participated in threat analysis, intrusion detection, and secure TCP/IP configuration to maintain 98% network uptime.

Python for Data Science, *Sphoorthy Engineering College, India* — **December 2019 – July 2020**

• Assisted in developing machine learning models, led EDA, enhancing expertise in SQL and statistical techniques for data-driven insights.
• Contributed to a 15% increase in user engagement through the development of a recommendation system.
• Demonstrated proficiency in end-to-end data science workflows, contributing to deliverables that informed data-driven decision-making.

Education

University of Dayton — **Dayton, OH, USA**
Master of Science in Computer Science — GPA: 3.74 | August 2025

Skills

Programming: Python, Java, C++, C#, SQL, JavaScript, PHP, HTML5, CSS, RESTful API Design
Frameworks: React, Django, Flask, Express.js, FastAPI
Cloud: AWS, Azure, GCP.
Database: PostgreSQL, MySQL, MongoDB.
DevOps: Docker, Kubernetes, Terraform, Jenkins, Git, Linux, Postman.
Certifications: AWS Academy Graduate – Cloud Architecting & Cloud Foundations, Cisco Networking Academy – Cybersecurity, Oracle Academy – SQL.

Projects

United Airlines Data Engineering Project

• Engineered a secure, scalable cloud ingestion pipeline on AWS, integrating S3, Glue, Redshift to process millions of flight records in real-time.
• Programmed and implemented robust ETL jobs in AWS Glue to cleanse, enrich, and normalize flight data, ensuring data integrity.
• Streamlined end-to-end pipeline management, from CSV ingestion to Redshift loading, using Step Functions, EventBridge and SNS based alerting.

Kafka AWS Stock Market Streamer

• Hosted Apache Kafka on AWS EC2 and crafted a Kafka Producer to stream simulated stock trading data from a CSV dataset to a Kafka topic.
• Spearheaded a Kafka Consumer to transform streaming messages into JSON format and stream the output into an AWS S3 Bucket.
• Consolidated AWS Glue Crawler to automatically catalog incoming data stored in S3, ensuring efficient data management.

Air Canvas: Mid-Air Drawing Using Computer Vision

• Built a real-time virtual canvas using Python and OpenCV by detecting fingertip movements via color segmentation and contour tracking.
• Granted capability for mid-air drawing and gesture-based mode switching (e.g., draw/erase) allowing users to sketch without physical contact.

Multitask Learning Model for Traffic Flow and Speed Forecasting

• Instigated machine learning model for vehicle detection, tracking, and counting using TensorFlow object detection API with SSD MobileNet.
• Annotated video frames and structured data saved in CSV format containing predictions of vehicle type, speed, direction, and color.
• Developed a system to classify vehicles, recognize color using KNN on color histograms, and estimate direction and speed via frame-wise pixel analysis.